



Tutorial 02

EE-809 Analog IC Design

DC I/V characteristics of MOS Transistors

Release: 1.0

Date: 14-Sept-2026



Copyrights ©, NUST Chip Design Centre (NCDC). All Rights Reserved. This document is prepared by NCDC and is for intended recipients only. It is not allowed to copy, modify, distribute or share, in part or full, without the consent of NCDC officials.



Contents

Introduction	4
Objectives	4
Cadence Environment Setup	4
Creating new Circuit Block.....	4
Schematic Capture	5
Simulation	6
Summary.....	15
Exercise Activites	16
Activity (1.1)	16
Activity (1.2)	16
Activity (1.3)	16
Activity (1.4)	16



Introduction

To effectively design a circuit, the behavior of the circuit elements must be known. This tutorial will give a walk-through in characterizing NMOS transistor using various Cadence tools. To make things concrete, in the end it will also be required to complete tasks in a similar way as will be shown in the following exercise steps.

Objectives

In the following exercise the aim of this tutorial will be to be able to draw a basic circuit, be comfortable with various cadence tools and to observe I/V characteristics of an NMOS transistor using TSMC PDK. In doing so, the following in the Cadence Virtuoso design suite will be explored.

- Schematic drawing and editing tool
- ADE (analog design environment) simulation tool
 - o DC analysis
 - o Parametric analysis
 - o Calculator tool

Cadence Environment Setup

After running Cadence Virtuoso, Library Manager/workspace window will popup. All the files appear on the first column, second column shows the circuit cells in the file and in the last column various attributes of the circuits appear like schematic, layout, symbol, maestro etc. The tutorial will begin with the first step of the design cycle of an electrical circuit; the schematic capture.

Creating new Circuit Block

1. Create new library (**File** → **New** → **Library**) and type a meaningful name for your library (like **Tutorial_02_yourname**).
2. A dialog box will appear, select **Attach existing Libraries** then attach PDK **tsmcN65**, into reference technology and click **Ok**.
3. In the library column select the file just created and go to file and create new Cell view



(**File** → **New** → **Cell View**), a dialog box will appear give a meaningful name (like **IV_CURVES_NMOS**) then click **OK**, new blank schematic editing window will appear.

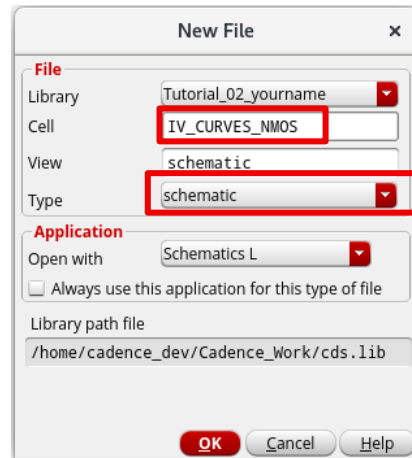


Figure 1: New Cell view windows setup

Schematic Capture

Schematic editor tool will be used heavily; using mouse to navigate across the tool might become cumbersome, it will be better to get used to the hot-keys to execute commands or functions.

4. In the schematic editing tool window, in the function ribbon click on (**Create** → **Instance**) to add a circuit component or use the hot-key "i"
5. **Add Instance** window will appear click on **Browse** and **Component Browser** window will appear
6. In the Component browser from library section find and click on **tsmcN65** then from Cell column find and click on **nch** or **nch_mac** and from the view section click on the symbol (note that on the Add Instance window various transistor parameters are shown. For now the default values will be used)
7. To place the component, click anywhere on the schematic editor window, copies of the symbol can be added until escape is pressed or by selecting another component from the component browser
8. For device characterization, use two **vdc** sources from the **analogLib** library: one for VGS (gate-to-source voltage) and the other for VDS (drain-to-source voltage), along with one **gnd (from analogLib)** for VSS.
9. For determining I/V characteristics varying VGS and VDS will be required; select a voltage source from the schematic window and right click, then click properties or use hot-key "q" to access component properties. In the **DC voltage** field type in **VGS** and leave rest of the fields as they are and click okay (note: The DC voltage value can also be fixed as 1V)
10. Similarly, do the same for the other voltage source with the variable **VDS**
11. It important that the circuit should look clean and show clear wire connections; from Edit pull-down menu click **rotate** or use "r" key to orient the component and they can also be flipped at different angles from the component properties



12. After placing components, next, go to **Create** → **wire** or use “w” to start placing wire connections between components. Simply click on the terminals of the components where wire needs to be placed
13. After making wire connection click save and read the status line to make sure there are no errors if there are errors, the error section will blink

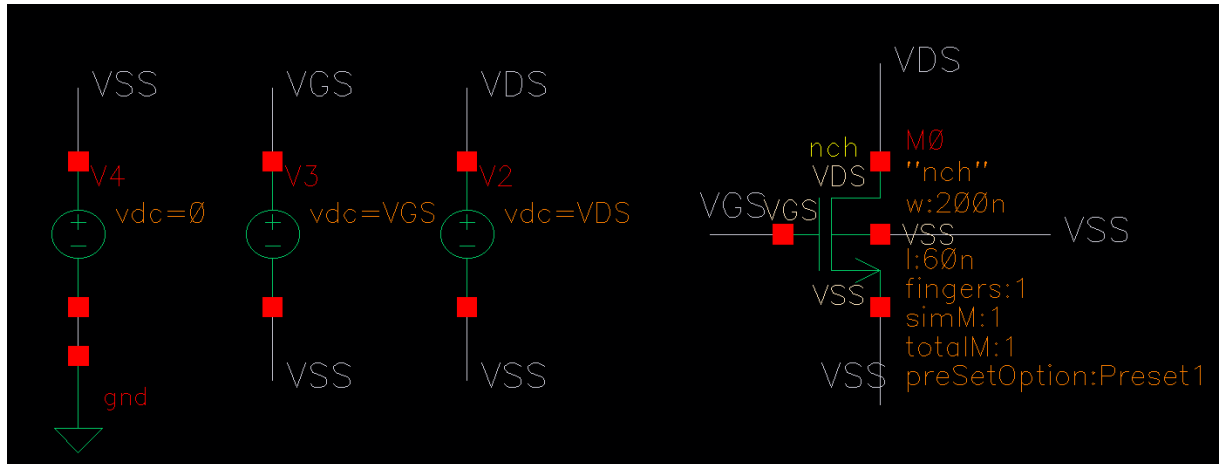


Figure 2: Circuit Schematic Drawn in the Editor

14. After completing the schematic, it should look similar to the one shown in Figure. 2

Simulation

After drawing the schematic, next, the circuit will be simulated to explore the behavior of the drawn circuit.

15. From the schematic window: **Launch** → **ADE Explorer**. In the launch ADE Explorer pop up, select “**Create New View**”.
16. On the **ADE** window right click on the **Design Variable** section, from the menu click **Copy from Cellview** and all variables assigned to various components will appear, double click on the empty field beside the variable and assign **2** to VDS and **1** to VGS (When entering values in Cadence Virtuoso, use SI prefixes only (e.g., m, u, n, p) and do not enter the units)
17. In the **ADE** window: **Analyses** → **Choose**; new window will appear, in the top section from where different analyses can be chosen, select **dc**, in **DC Analysis** section check mark the **Save DC Operation Point**, in **Sweep Variable** section check **Design Variable** and a new tab will appear, click on the **Select Design Variable** a new window will appear and from the list choose **VGS**, in **Sweep Range** section choose **Start-Stop** and specify range as **0 to 1**, then click **OK**

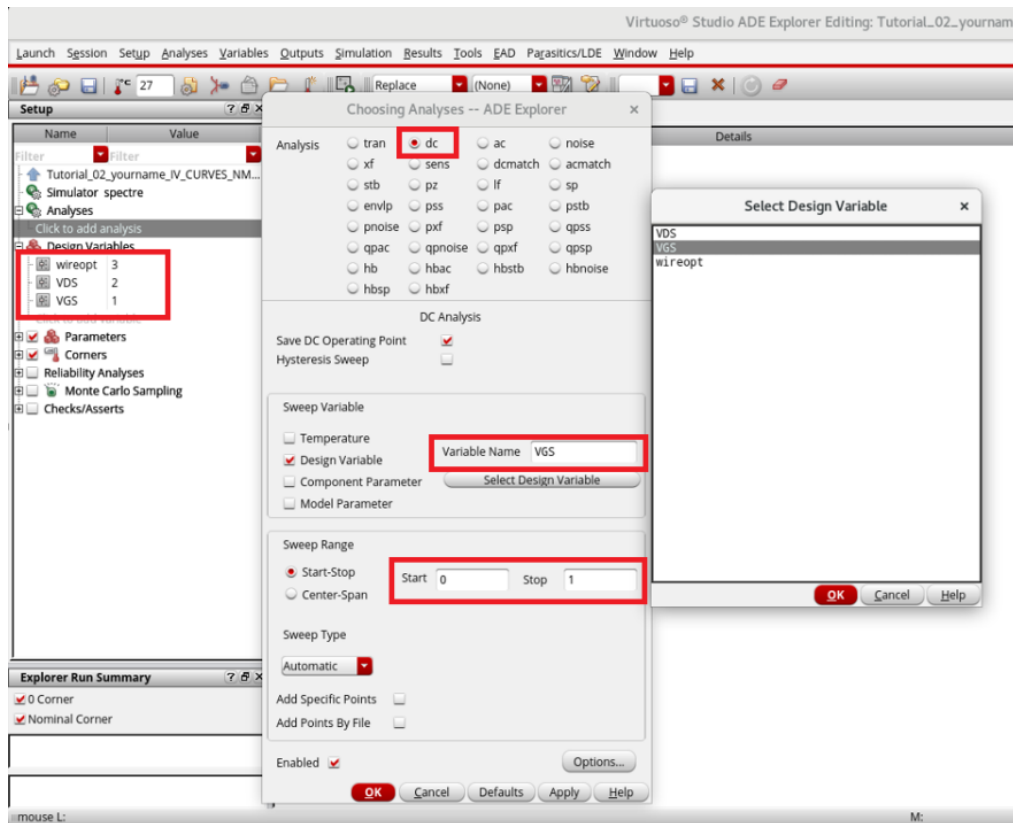
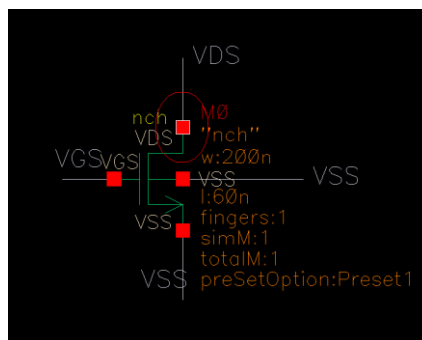


Figure 3: Choosing Analyses and ADE window

18. Now the outputs to be observed will be set, on the **ADE window** on the pull-down menu: **Outputs** → **To Be Plotted** → **Select On Design**, on the schematic editor window click on the drain terminal (the one shown in red box) of the transistor (not the wire) and a circle will appear on the schematic, as shown in the figure below.



19. Then go back to the **ADE window** and verify if the drain current of the MOSFET has been added.
20. On the right side of the ADE window, find **Run Simulation (Green Triangle)** button and click, **Id versus VGS graph** will appear with **constant VDS**



-

23. The operating-point data provides various parameters of the MOSFET, such as its region of operation, Early voltage, and other device characteristics.

$$gm = \frac{\partial I_D}{\partial V_{GS}} \mid V_{DS} \text{ is constant}$$

8



24. To perform mathematical operations on the graph, tool of ADE called **Calculator** will be used, Calculator can be accessed either from the ADE window or the graph display window: **Tools** → **Calculator**, when the calculator window appears, above the text field select **Wave** and click the trace on the graph, the graph equation will appear in the calculator text field, below the text field, there are list of mathematical functions which can be performed on the equation, click on the pull-down menu above the list and select All, from the list find the function **deriv** and select then click **Evaluate** button situated above the text field, a graphical trace will be generated which in this case will be the transconductance curve.

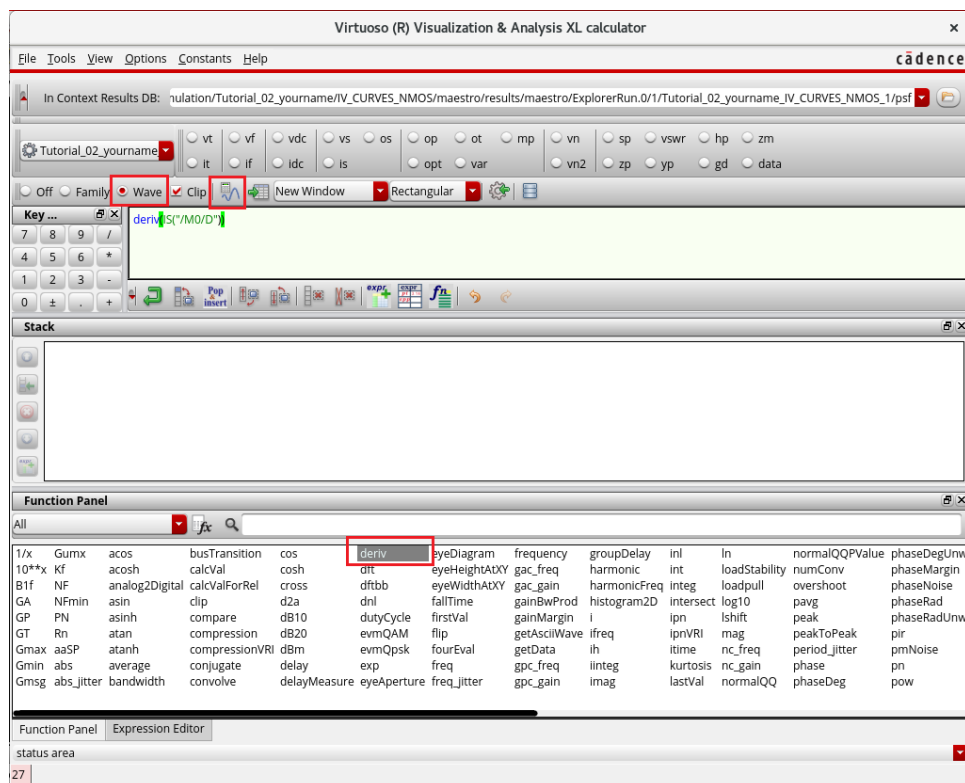


Figure 5: Calculator window



25. Step 24 is quick way of getting results, but things may get cumbersome when graphs need to be calculated after every run, you can resolve this by saving the equations in the output section of the ADE, follow the steps in **step no. 24**, then once the expression is formed in the calculator, click on the icon as shown in the picture, it will send the expression to the ADE window. Under the name column, name this new expression as **gm**

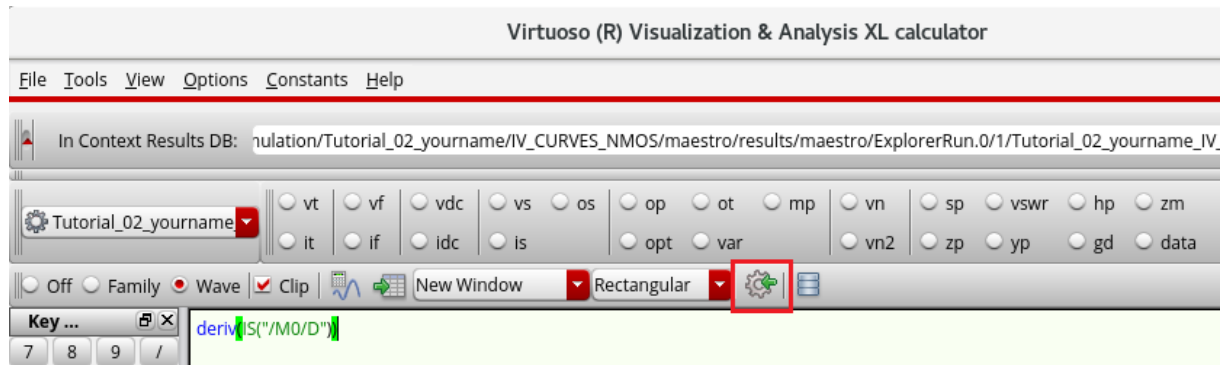


Figure 6: Output Setup window

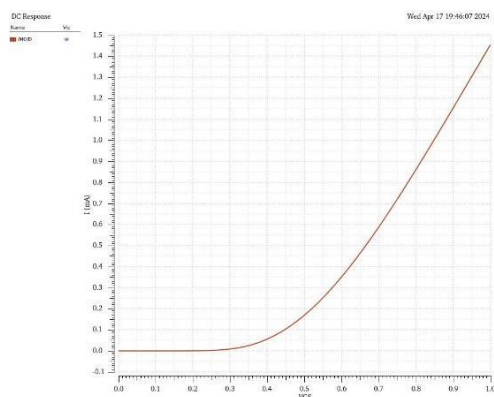


Figure 7: Id vs. Vgs

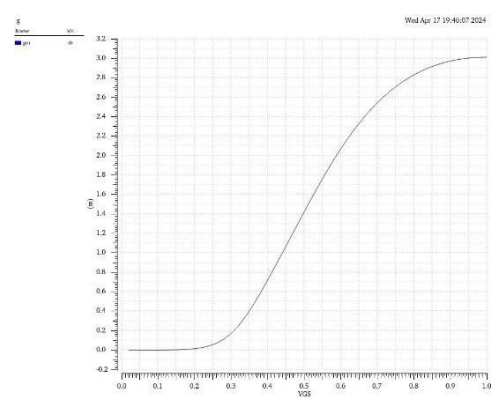


Figure 8: Derivative of Id curve

Next, is to determine I_D vs. V_{DS} with varying V_{GS} . Note that now two variables need to be varied or swept. For this **Parametric Analysis** will be used.

26. On the ADE window double click on **DC analysis** in the Analyses section, , change the design variable to **VDS (sweep range as 0 to 2)** and keep the rest of the setting as they are and click **OK**
27. On the ADE window, in the outputs section, uncheck all output plots except M0/D.
28. Click **Netlist and Run (under Simulation tab)** or **press on the green triangle on the right side**, resulting graph will appear (Note that graph should reconcile since VGS was kept as fixed (VGS= 1) in the Design variable section).
29. To see the plot of ID v/s VDS for more than one value of VGS, go to design variable section on the left side under the analyses section and double click on the space where its value is specified (1 in this case). Now click on the three dots, a new window called **“parameterize”** will open. In this window select **From/To** from the top right section called “Add Specification”.
30. Set the range from **0 to 2** and Step Mode to **Linear Steps** with **0.4** Step Size

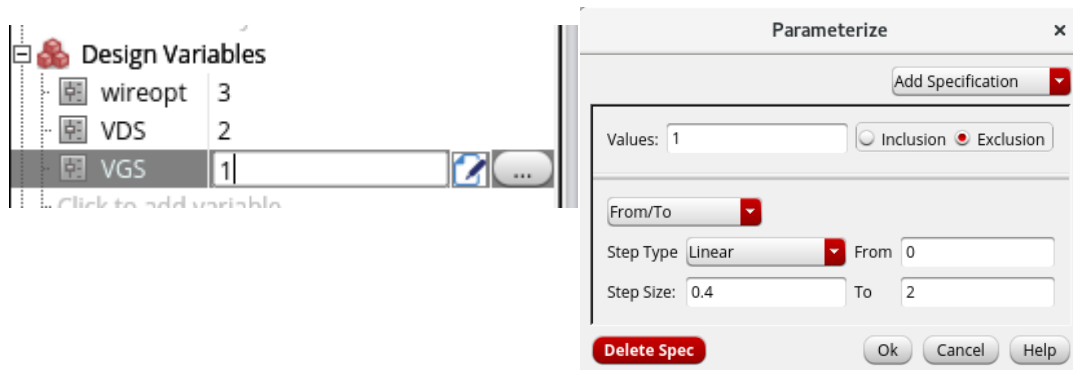


Figure 9: Parametric Analysis window

31. Now run the simulation and the curves of ID v/s VDS for multiple VGS values will be plotted.
32. Note that, we expected that current should stay constant in the saturation region, but it doesn't and keeps increasing. This is due to a second order effect called **channel-length modulation**

Notice that the variable set in **Choosing Analyses** window will automatically be set as independent variable (set on x-axis).

33. With parametric analysis, two variables were varied which in turn also effected the transistor device parameters, among many parameters, we can observe transconductance yet again from ADE: **Tools** → **Results Browser**, a new window will appear, from **Signals** tab click on **dcOpInfo-info** then click **M0**, a list of device parameters will appear, double click **gm** and a (familiar) graph of transconductance will appear.

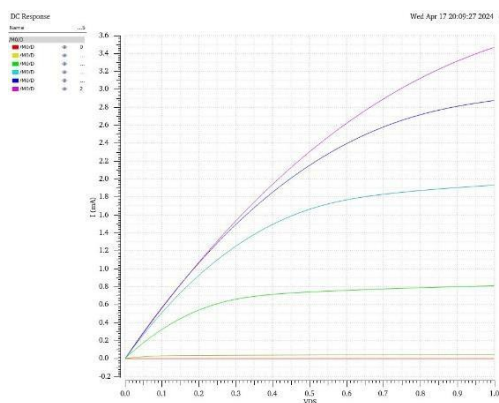


Figure 10: I/V curve for

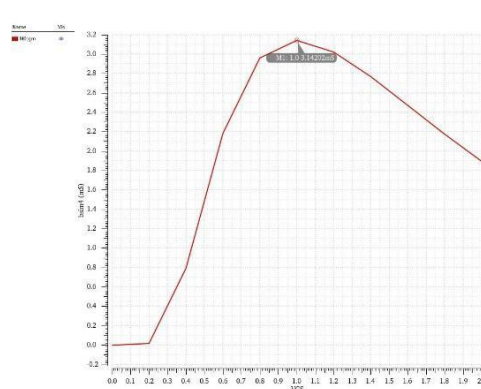


Figure 11: Transconductance from Results Browser



34. Standard long-channel theory assumes a MOSFET shuts off abruptly when V_{GS} falls below V_{TH} . In real silicon, however, the transition is gradual: near $V_{GS} \approx V_{TH}$, a weak inversion layer remains active, giving rise to **subthreshold conduction** where current flows from drain to source. To observe this effect, plot the drain current vs varying gate to source voltage with a fixed drain to source voltage as done before.
35. In cadence v the regions of the MOSFET are denoted by numbers.
- Region 0** = Cut Off
- Region 1** = Triode/linear region
- Region 2** = Saturation region
- Region 3** = Subthreshold conduction region
36. To observe this effect, after plotting the drain current, plot the regions of operation along with the drain current. To plot regions, a simulation file needs to be provided to the tool so it can save all DC operating points which includes regions plot. To do this first click on **Activities** on the top left corner, then click on the **files** icon. Open your work directory (SEECs Folder). From here right click and click on **Open in Terminal**. From terminal type the following command: **nano save_operating_points.scs** and hit enter. Now an empty text editor will open. Here write this line **save M0:all** and then press **ctrl +s** and close this terminal. This will create text file. Here M0 denotes the instance name of the MOSFET in the schematic. If the instance name in schematic is for example M1, type M1 instead of M0.
37. Now go back to Virtuoso by using the activities tab from the top left corner.
38. In the **ADE** window go to **Setup** then go to **simulation files**. In the window that opens, under **Definition files** import the text file just made.

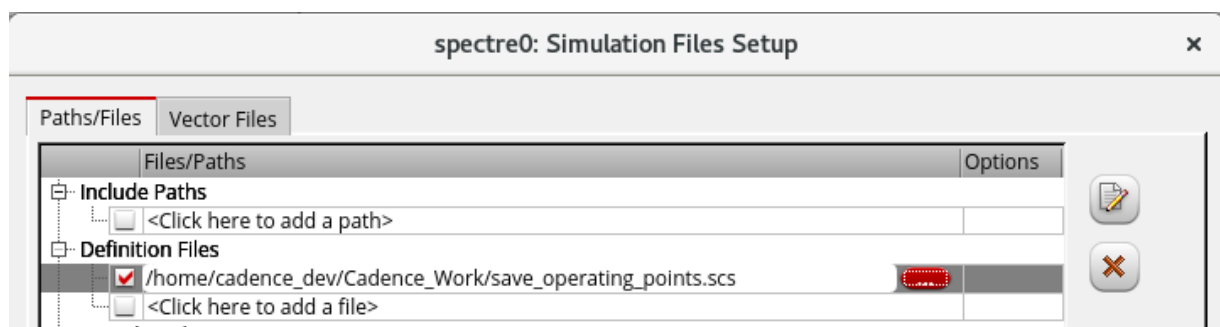


Figure. 12 Simulation file setup

39. After the file is imported, click **OK**.
40. Now run the simulation for drain current versus the gate to source voltage simulation (ID V/S VGS). After simulation is done, go to results browser, and under the **dc**, look for the variable called **M0:region**



Figure 10 is a graph showing the current (I) in microamperes (uA) versus the voltage (V) in Volts (V) for the M1-M9 regions. The x-axis ranges from 0.0 to 1.0 V, and the y-axis ranges from 0.0 to 100.0 uA. The M1-M9 regions are highlighted in green, and the M10-M12 regions are highlighted in orange. The M1-M9 regions show a sharp increase in current at low voltages, while the M10-M12 regions show a gradual increase in current across the entire voltage range.

Region	V (V)	I (uA)
M1	0.120.0m	3.0
M2	0.369.0m	3.0
M3	0.380.0m	2.0
M4	0.380.0m	2.0
M5	0.100.0m	0.0
M6	0.100.0m	1.63243nA
M7	0.120.0m	2.81837nA
M8	0.120.0m	2.81837nA
M9	0.380.0m	1.41332uA

Figure. 14 Subthreshold conduction region

- $$V_{TH} = V_{TH0} + \gamma(\sqrt{|2\phi_F| + V_{SB}} - \sqrt{|2\phi_F|})$$

- 13



under dc look for **M0:vth**. Plot the threshold voltage and send it to ADE.

45. Remove the VSS label from the bulk and use a dc voltage source as shown below.

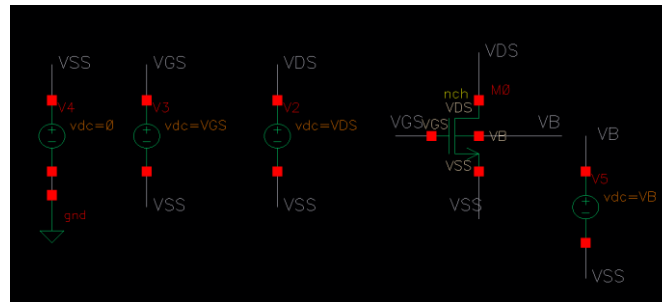


Figure.15 Schematic for body effect simulation

46. Next go to ADE explorer/maestro window, and from **design variables** right click and select **copy from cellview**. After VB is imported to design variable, assign the value of 0 to it. Next go the **Analyses** tab and select **Choose**, select **dc** and in the design variable section go to **Select Design Variable** and select VB from the list. In the Sweep Range, specify a range of -1 to 0. Plot as shown below will be plot showing that as VB goes below the source voltage, which is kept at VSS, the threshold voltage increases.

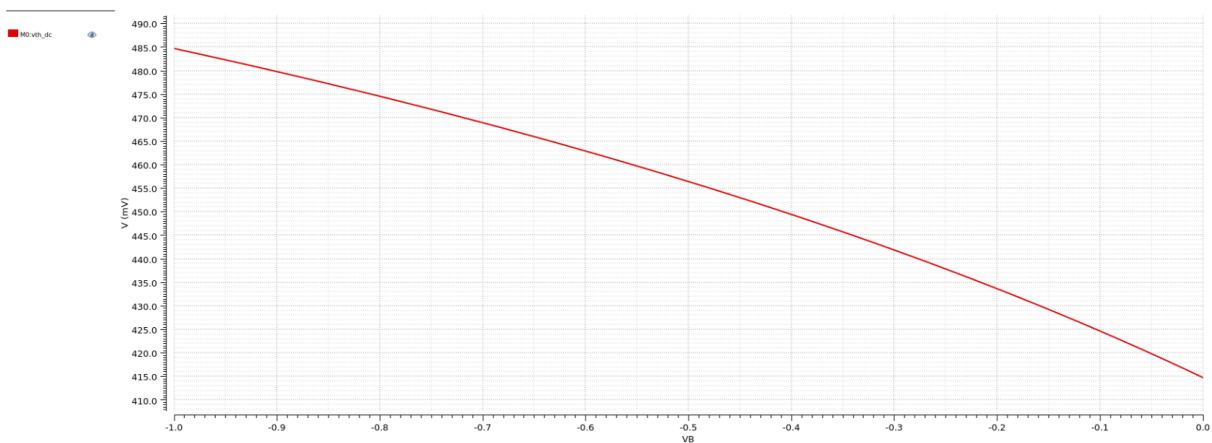


Figure.16 Plot of threshold voltage against changing body voltage.



Summary

- Began with setting up file system for the circuit block
- Schematic editing tool was then explored and drew a circuit for characterization
- Explored the ADE for DC analysis of the circuit and schematic node annotations
- Used Calculator tool for post-simulation analysis
- Learned to use parametric sweep tool to vary more than one variable
- Explored the subthreshold conduction region by examining the MOSFET's drain current alongside its operating region
- Investigated the body effect by observing how lowering the bulk voltage increases the MOSFET's threshold voltage.



Exercise Activities

Activity (2.1)

- Repeat the exercises above for PMOS transistor and generate the following graphs

I. I_D v/s V_{GS} or I_D v/s V_{SG} (for a constant V_{DS} or V_{SD}).

Activity (2.2)

- Repeat the exercises with PMOS transistor and generate the following graphs

I. I_D v/s V_{DS} or I_D v/s V_{SD} (for a constant V_{GS} or V_{SG})

II. Perform parametric analyses and generate I_D v/s V_{DS} or I_D v/s V_{SD} (for multiple V_{GS} or V_{SG})

For the I_D v/s V_{DS} graphs explain why the curve does not remain constant after V_{DS} crosses the overdrive voltage. Explain this effect. What could be done to make the curve as straight as possible. Support your solution, if any, with simulation results clearly showing before and after results.

Activity (2.3)

- Show the subthreshold conduction region for a PMOS device just as done with the NMOS device.

Activity (2.4)

- How does the threshold voltage of the PMOS transistor vary with body voltage. Support your answer with simulation results.
- Investigate the effect of temperature on threshold voltage for a PMOS transistor. Support your answer with simulation results. (Hint: temperature can be varied from **analyses-> dc**. Assume a temperature range of -40 to 125 °C)